

ORIGINAL RESEARCH

A Coordinated Community- and Family-Based Intervention to Control Blood Pressure in Rural China

A Cluster-Randomized Trial

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ABSTRACT

BACKGROUND Current hypertension prevention strategies predominantly focus on identifying and treating individuals with established hypertension. Whether a coordinated community- and family-based intervention can shift the population distribution of systolic blood pressure (SBP) remains uncertain.

OBJECTIVES The purpose of this study was to evaluate whether a coordinated, community- and family-centered intervention could shift the distribution of SBP at the population level in rural China.

METHODS In this cluster-randomized trial conducted in rural China, village clusters comprising adults aged 40 to 80 years, irrespective of blood pressure (BP) level or hypertension status, were randomly assigned (1:1) to a multifaceted intervention or usual care. The intervention included community instructor support, potassium-enriched low-sodium salt substitute, home BP and weight monitoring, physical activity promotion, and facilitation of antihypertensive treatment during an initial 6-month active intervention phase, followed by continued digital BP monitoring support through 12 months. Office BP was measured at baseline and at 6 and 12 months by trained staff independent of the intervention team. The primary outcome was the between-group difference in change in SBP from baseline to 6 months. Analyses followed the intention-to-treat principle using linear mixed-effects regression models.

RESULTS A total of 80 village clusters, comprising 8,001 participants, underwent randomization. Baseline mean SBP was 133.1 mm Hg in the intervention group and 131.4 mm Hg in the control group. At 6 months, mean SBP fell by 6.0 mm Hg in the intervention group and increased by 5.1 mm Hg in the control group, resulting in an adjusted between-group difference of −10.7 mm Hg (95% CI: −11.8 to −9.6 mm Hg). BP <130/80 mm Hg was achieved in 46.2% of participants in the intervention group vs 23.6% in the control group (adjusted OR: 5.3; 95% CI: 4.3–6.6). At 12 months, the adjusted between-group difference in SBP was −3.7 mm Hg (95% CI: −4.9 to −2.6 mm Hg). Serious adverse events occurred in 2.8% of participants in the intervention group and 2.3% in the control group.

CONCLUSIONS In rural China, a coordinated community- and family-based intervention significantly shifted the population distribution of SBP toward lower levels over 6 months. Attenuation of the treatment effect over time suggests that sustained population BP reduction may require ongoing implementation infrastructure and continued community support. (Effect of Healthy Family Program on Population Blood Pressure; [NCT06427096](https://doi.org/10.1016/j.jacc.2026.07.045)) (JACC. 2026;■:■–■) © 2026 by the American College of Cardiology Foundation.

**ABBREVIATIONS
AND ACRONYMS****AESI** = adverse events of special interest**BP** = blood pressure**DBP** = diastolic blood pressure**GPAQ** = Global Physical Activity Questionnaire**Mini-EAT** = Mini-Eating Assessment Tool**RRHI** = Ruyang Rural Health Institute**SAE** = serious adverse event**SBP** = systolic blood pressure

Blood pressure (BP) is continuously associated with cardiovascular risk across populations, without a clear biological threshold below which risk is absent.^{1,2} Accordingly, hypertension represents an arbitrarily defined point along a continuous distribution of BP, and the proportion of individuals classified as hypertensive is closely related to the population mean BP level.³ Shifting the population BP distribution toward lower levels, therefore, is expected not only to reduce overall cardiovascular risk in the population, but also to reduce the prevalence of hypertension.⁴ Nevertheless, contemporary

hypertension management remains largely clinic-centered and diagnosis-driven, focusing predominantly on the pharmacologic treatment of individuals with established hypertension within health care settings.^{5,6}

Although moderately effective for high-risk patients, this individual-focused and treatment-oriented approach does not adequately address the broader social, behavioral, and environmental determinants that shape population BP distribution and contribute to the development of hypertension.^{4,7} Moreover, as most cardiovascular events arise from the large segment of the population with normal or mildly elevated BP rather than the smaller group with markedly elevated BP, exclusively clinic-centered approaches focusing on high-risk individuals may have limited impact on the overall population cardiovascular burden. Because cardiovascular risk increases continuously across the BP spectrum, even modest leftward shifts in the population distribution of BP have the potential to prevent large numbers of cardiovascular events.⁷ Accordingly, strategies that extend prevention beyond the clinic and into community settings may complement traditional diagnosis- and treatment-based approaches by addressing upstream determinants of BP at scale.

The limitation of a predominantly clinic-based, individual-level prevention model is particularly salient in low-resource settings, where access to health care and primary care capacity remain constrained. In China, village doctors play a central role in hypertension management, particularly in rural areas.⁵ However, the limited primary care workforce, with only 2.8 primary care physicians per 10,000 population,⁸ restricts the scalability of clinic-based prevention strategies in these settings. Accordingly, community- and family-based interventions focusing on dietary patterns, physical activity, health care utilization, and medication adherence—factors that are strongly shaped within family and community environments—has the potential to shift the population distribution of BP while simultaneously improving BP control among individuals with established hypertension.

We conducted the Healthy Family Program, a cluster randomized controlled trial designed to evaluate whether a coordinated community- and family-based intervention could reduce BP among adults aged 40 years or older at the population level, regardless of their baseline hypertension status.

METHODS

TRIAL DESIGN AND OVERSIGHT. The trial was conducted in Ruyang County of Henan Province, a county with a population of 434,770 and an economic profile representative of rural regions in China ([Supplemental Tables 1 to 5](#), [Supplemental Figure 1](#)). Given that the county is located in northern China, the average maximum temperature in Ruyang County typically reaches 35 °C in the summer and drops to below 10 °C in the winter ([Supplemental Figures 1 and 2](#)). Moreover, in the winter, there is no continuous heating system in houses to maintain a constant indoor temperature.

Preparations for the trial were undertaken between March 2024 and April 2024, and this included

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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meetings with village leaders and community groups to inform them about the conduct of the study. During the preparation period, a community health instructor was identified in each village to receive training on the study protocol and to screen eligible families for participation. The health instructors, typically members of the village Women's Federation with experience in family planning, possess basic health knowledge and strong skills in engaging villagers. With China's relaxed family planning policy, they are well-positioned to shift their focus to health instruction, leveraging their expertise to promote community well-being.

The design was an open-label, parallel-group, cluster-randomized controlled trial. The unit of randomization was the cluster (village) and the unit of effectiveness analysis was participants living in the clusters. Details of the rationale, design, and analysis of the trial protocol have been published⁹ and are available in the [Supplemental Appendix](#). Funding was through a grant from the Non-communicable Chronic Diseases-National Science and Technology Major Project (grant number: 2024ZD0526800/2024ZD0526805) and the International Exchange and Cooperation Center of the National Health Commission of the People's Republic of China. The study was approved by the Ethics Committee of Ruyang County People's Hospital, an appropriately constituted and regulated Institutional Review Board that adheres to national guidelines for human subjects research (March 30, 2024; approval number: RY-20240330-01). The project was registered in the National Medical Research Registration Information System and subject to ethical oversight by the China Clinical Trial Registry Center. Written informed consent was obtained from all participants. The Ruyang Rural Health Institute coordinated the trial in accordance with the principles of the Declaration of Helsinki and guidelines for Good Clinical Practice and performed the analysis. An independent Data and Safety Monitoring Board was not convened because of the pragmatic design of the study, the low-risk nature of the intervention, and the rapid completion of participant recruitment and follow-up, which made interim safety review impractical. Safety was monitored throughout the study by the site research staff through the prospective collection and documentation of adverse events in accordance with the study protocol.

ELIGIBILITY. Villages were eligible for inclusion if they contained at least 200 permanent households, including a minimum of 60 households with 2 or more family members aged 40 to 80 years. Eligible

households were required to have at least 2 members in this age range who met all individual eligibility criteria, and at least 1 household member (regardless of participation in the trial) capable of using a smartphone to record health data (eg, BP, weight). Participants were adults aged 40 to 80 years, irrespective of their baseline BP level or hypertension status, who had no severe cognitive impairment, advanced malignancy, requirement for ongoing dialysis, other serious diseases or physical disability, nor any plans for travel exceeding 1 month during the study period or pregnancy in the next 6 months. A complete list of the eligibility criteria is available in the study protocol.⁹ We did not include individuals younger than 40 years, as they are more likely to leave rural villages for work or study, making them less available for receiving the intervention and for follow-up.

RANDOMIZATION. Randomization occurred after all villages had identified 30 to 50 eligible households, with approximately 100 participants per village, ensuring that participant eligibility was determined prior to allocation and thereby minimizing recruitment bias. Villages were stratified according to their administrative township (7 in total) and then randomly assigned in a 1:1 ratio to receive either the healthy family intervention or usual care (control). The Ruyang Rural Health Institute (RRHI) conducted the randomization process and kept this secure from participants. Researchers, community health instructors, villagers, and village doctors were all informed of the group assignments, but the study statistician was kept blind to these details.

INTERVENTION. Full details of the intervention are available in the trial protocol,⁹ with the key components summarized below and in the [Central Illustration](#).

The first component was establishing a BP management team led by community health instructors. The trained team comprised community health instructors (village Women's Federation members with health training), family leaders (designated members responsible for promoting health behaviors within their households), and village doctors (rural health care professionals providing basic medical and public health services). Health instructors received modest incentives that were conditional on adherence to protocol requirements, with deductions applied for noncompliance.

The second component involved multifaceted interventions, which comprised 6 evidence-based components. 1) Lifestyle education: community health instructors received initial training at baseline

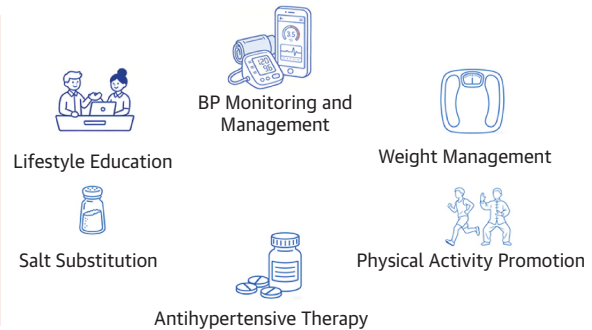
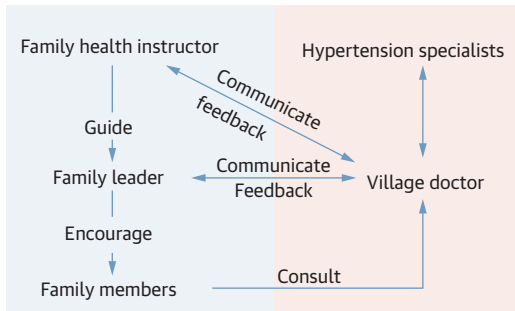
CENTRAL ILLUSTRATION Design and Key Findings of the Healthy Family Program**Healthy Family Program**

- Multicenter, parallel-group, open-label, cluster-randomized controlled trial
- 80 villages of Ruyang County, Henan Province, China
- 8,001 participants aged 40-80 years old, irrespective of baseline blood pressure

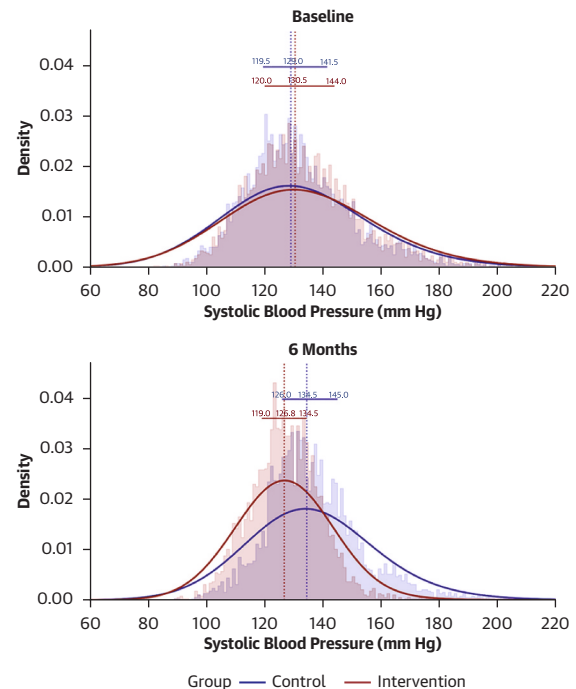
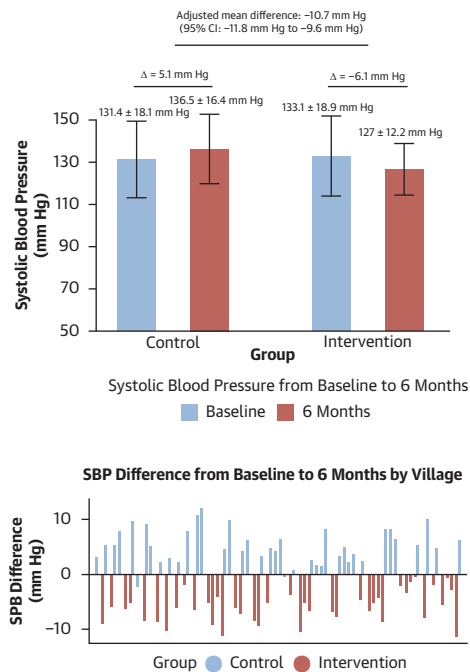
Intervention

1) Establish a BP management team led by community health instructors

2) Implement multifaceted interventions

**Key Results**

- At 6-months, the mean SBP fell by 6.1 mm Hg (to 127.0 mm Hg) in the intervention group and increased by 5.1 mm Hg (to 136.5 mm Hg) in the control group
- The mean adjusted reduction being 10.7 mm Hg greater with the intervention



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In this cluster-randomized trial conducted in 80 rural villages in China, 8,001 adults aged 40 to 80 years were assigned to a coordinated community- and family-based intervention or usual care. The intervention integrated community health instructors, village doctors, family engagement, home BP monitoring, lifestyle education, salt substitution, physical activity promotion, weight management, and antihypertensive treatment support. At 6 months, mean systolic blood pressure decreased by 6.1 mm Hg in the intervention group and increased by 5.1 mm Hg in the control group, resulting in an adjusted between-group difference of -10.7 mm Hg. Density plots demonstrate a leftward shift in the population distribution of systolic blood pressure associated with the intervention.

and refresher training at month 4 at the coordinating center, after which they provided education sessions at a central village office for the participating family leaders (approximately 50 per village) on salt reduction, weight management, BP control, and physical activity. All households received educational materials, and village-based social media WeChat groups enabled ongoing health communication. 2) Salt substitution: a free monthly supply of potassium-enriched salt substitute (400 g per member, composition of 88% sodium chloride and 12% potassium chloride by mass) was provided. Participants were instructed to use it for all home cooking and food preparation. 3) Weight management: each household received a free digital scale and was instructed to record weights via a customized WeChat mini-program, with the requirement that family leaders upload these data at minimum monthly intervals. 4) Physical activity promotion: community health instructors were encouraged to organize group exercise sessions 3 to 5 times per week. These sessions, which included activities such as tai chi and brisk walking, were to last 45 to 60 minutes. 5) BP monitoring and management: each household received a digital BP monitor (Yuwell, YE690D) and was instructed to perform standardized twice-daily measurements (after a 5-minute rest, duplicate readings 1 minute apart) following the 2019 Chinese family BP monitoring guideline.¹⁰ Readings were manually entered into the WeChat mini-program. The system automatically calculated daily average BP, applied a color-coded classification, and sent corresponding short message alerts. These messages notified participants of their next measurement date and delivered tiered clinical advice: seek review within 1 week if BP was persistently elevated (mean $\geq 140/90$ mm Hg) or seek immediate care for critical values (SBP >180 / <80 mm Hg or DBP >120 mm Hg). Missed uploads triggered automated reminders to the family leaders. Community health instructors monitored village-wide data via the WeChat mini-program, contacting families when necessary. 6) Antihypertensive therapy: family leaders and health instructors were encouraged to ensure participants with sustained high BP according to the Chinese guidelines ($\geq 140/90$ mm Hg)^{11,12} sought medical treatment from village doctors, with specialist consultations organized if BP was not well controlled for ≥ 3 months.

The intervention began after baseline data were collected and continued for 6 months, after which feedback from the coordinating center, incentives to health instructors, and distribution of free salt substitute ceased. The BP and weight monitoring devices

and digital platform remained active throughout the study.

Participants in the control group received usual BP management through routine local healthcare services and were not provided with BP monitors, access to the digital support platform, lifestyle education, or low-sodium salt.

PROCEDURES. Data were collected concurrently across both intervention and control groups by trained research coordinators at a central village office at baseline, and at 6- and 12-month follow-up visits. Intervention delivery personnel were not involved in the data collection process, which included demographic characteristics, medical history, lifestyle factors, and the medications of participants. Physical activity was assessed using the World Health Organization's Global Physical Activity Questionnaire (GPAQ),^{13,14} which consists of 16 questions to assess an adult's level of physical activity across multiple domains (work, moving between places, and leisure activities) and sedentary behavior in a typical week. Dietary habits were assessed using the Mini-Eating Assessment Tool (Mini-EAT),¹⁵ a short 9-item questionnaire to assess consumption of common food groups known to affect health. BP was measured using a BP monitor (Microlife, B2 Basic) in a sitting position, with a cuff suitable for arm circumferences ranging from 22 to 42 cm. No participant was observed to fall outside this recommended cuff range. After a 5-minute rest period, 3 BP measurements were taken in a single session, with a 1-minute interval between each reading,¹⁰ with the mean of the second and third measurements used for analysis.

At the 6-month follow-up visit, data were obtained on any cardiovascular disease events and serious adverse events (SAEs) according to standard definitions, and the adverse events of special interest (AESIs) of hypotension, syncope, hyponatremia, hyperkalemia, injurious falls, and acute kidney injury.

OUTCOMES. The primary outcome was the change in SBP from baseline to 6 months, comparing the intervention and control groups. Secondary outcomes included: the change in SBP between the groups at 12 months; change in diastolic blood pressure (DBP) between the 2 groups at 6 months; the proportion of participants with an SBP <130 mm Hg and DBP <80 mm Hg at 6 months; and proportion of participants receiving antihypertensive therapy at 6 months. Safety outcomes were cardiovascular events, all-cause mortality, SAEs, and AESIs during the study period. The costs associated with implementing the intervention were also evaluated.

Descriptive analyses of GPAQ, Mini-EAT, weight, medication use, and self-monitoring platform use during the trial period (6 months) and extension to 12 months were also performed.

STATISTICAL ANALYSIS. A total of 80 clusters each with 100 participants was estimated to achieve 80% power to detect a 3 mm Hg difference in the change in SBP between the intervention and control groups, with an assumed SD of 19 mm Hg, an intraclass correlation coefficient of 0.05, and a 2-sided type I error rate of 5%. The statistical power was calculated using a 2-sample *t*-test for cluster-randomized trials with PASS software version 15.0.5.

All analyses were conducted according to the intention-to-treat principle at the participant level.¹⁶ The primary and secondary outcomes were analyzed using a linear mixed-effects regression model, with township, age, sex, baseline SBP, and hypertension status included as fixed effects, and village and family (nested within village) as random intercepts. Village-level intraclass correlation coefficients (ICCs) were estimated to account for correlation among participants within the same village. To account for potential variation in visit timing, we performed a sensitivity analysis incorporating actual visit week as a time-varying covariate in the model. Fixed effects included intervention group, visit, and the interaction between intervention group and visit, with random intercepts for villages. A compound symmetry covariance structure was specified for within-participant repeated measures over time. Adjusted mean differences (MDs) and corresponding 95% CIs were derived from the fitted model. All secondary outcomes were prespecified with multiplicity adjustment using a sequential Holm-Šidák procedure. Sensitivity analyses were performed at both the village and family levels. No imputation was conducted in the primary analysis due to minimal missing data. Baseline characteristics of participants with and without 12-month follow-up data were compared by treatment group to evaluate potential bias related to missingness. Sensitivity analyses were subsequently performed using multiple imputation under a missing-at-random assumption. Missing 12-month SBP values were imputed using chained equations across 20 datasets, with estimates combined according to Rubin's rules. The distribution of SBP was presented using a histogram with a bin width of 1 mm Hg, and density curve derived using kernel density estimation (KDE).¹⁷ Prespecified subgroup analyses were conducted by age, sex, education level, history of hypertension, baseline SBP, baseline DBP, baseline Mini-EAT scores, family size,

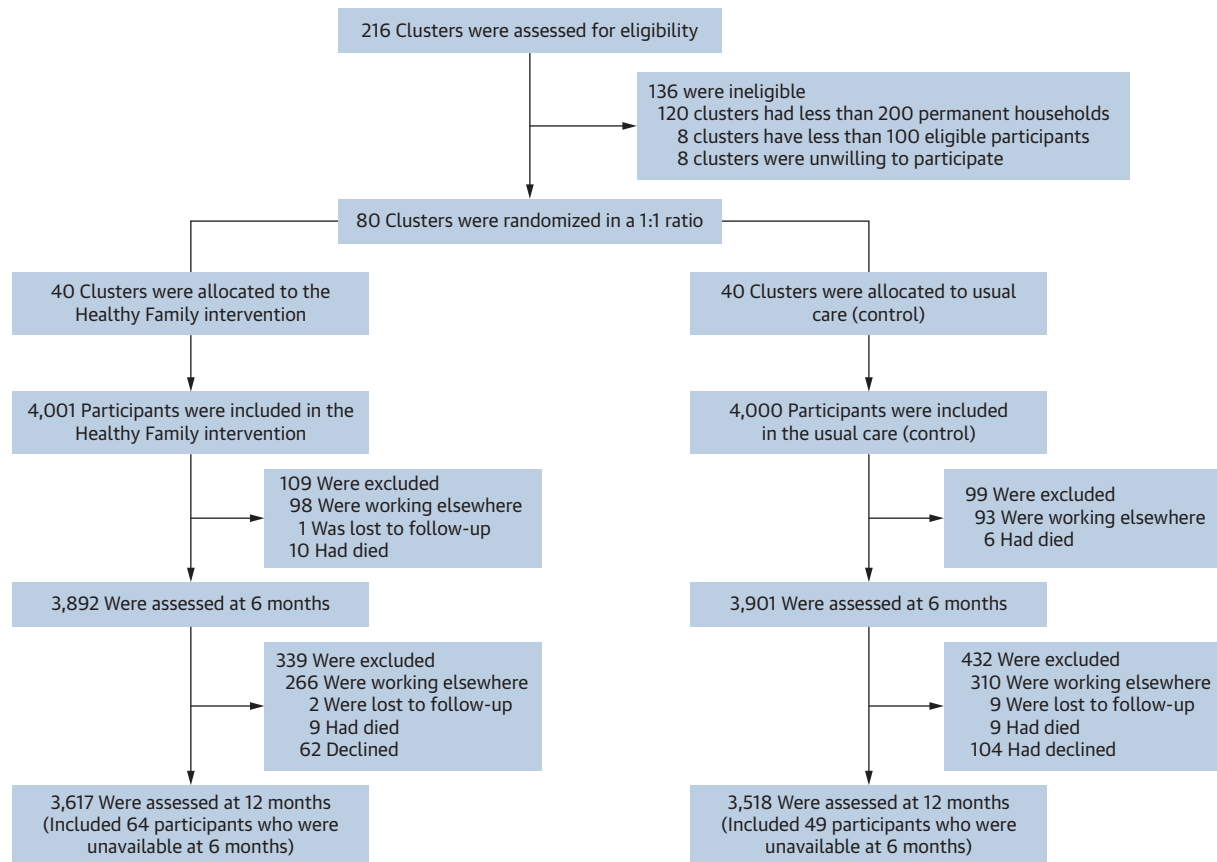
and township socioeconomic status. An additional exploratory post-hoc subgroup analysis was performed according to baseline antihypertensive medication use. The small number of cases did not permit an analysis by history of cardiovascular events. Treatment-by-subgroup interaction coefficients were evaluated in the primary model to assess heterogeneity of treatment effects. Post hoc analyses included: description of SBP difference from baseline to 6 months by village; medication type among patients with hypertension at 6 months, evaluated using a chi-square test; and WeChat platform usage status in the intervention group. All statistical analyses were performed using R version 4.5.1 and SAS version 9.4.

The incremental cost of delivering the intervention was prospectively estimated using an activity-based costing approach that accounted for all nonrecurring costs, including labor, materials, supplies, and services required for implementation.

RESULTS

PARTICIPANTS AND TRIAL COHORTS. From November 2023 to March 2024, all 216 villages in Ruyang County were screened for eligibility. After excluding 120 villages that did not meet the prespecified criteria, the remaining 96 villages were invited to participate, of which 80 agreed and were randomized equally ([Figure 1](#)). Village and household characteristics were reasonably balanced between randomized groups ([Supplemental Table 6](#)). The mean age of the 40 health instructors was 47.6 ± 8.3 years and 100% were female. There were 1,965 family leaders assigned, with a mean age of 55.1 ± 10.7 years and 42.3% were female ([Supplemental Table 7](#)).

Baseline data collection in 4,001 participants in the intervention group and 4,000 participants in the control group occurred in the summer, between May 23, 2024 and July 18, 2024. The 6-month follow-up assessments were undertaken in the winter, between November 18, 2024, and February 12, 2025 ([Supplemental Figure 2](#)). Of these, 3,892 participants (97.3%) in the intervention group and 3,901 participants (97.5%) in the control group completed their 6 month follow-up assessments. The 12-month assessments occurred between May 14, 2025 and July 27, 2025. A total of 3,617 participants (90.4%) in the intervention group and 3,518 participants (87.9%) in the control group completed the 12-month follow-up data collection. Baseline characteristics of participants were well-balanced between the intervention and control groups. The mean age was 61.2 ± 9.2 years, 51.4% were female, 9.8% had diabetes, and

FIGURE 1 Enrollment, Randomization, and Follow-Up of Participants

0.3% had chronic kidney disease (Table 1, Supplemental Table 8). There were minor differences in the hypertension status of participants at baseline: a history of hypertension or BP $\geq 140/90$ mm Hg occurred in 36.1% and 30.8%, and mean SBP was 133.0 ± 18.9 mm Hg and 131.3 ± 18.1 mm Hg, in the intervention and control groups, respectively (Table 1). There were no major between-group differences in characteristics by age and sex (Supplemental Tables 9 to 12), nor in the awareness of existing hypertension, use of antihypertensive treatment, and level of BP control ($<130/80$ mm Hg) among participants (Supplemental Table 13). Baseline characteristics were broadly comparable between participants with and without 12-month follow-up data in both treatment groups (Supplemental Tables 14 and 15).

PRIMARY EFFICACY RESULTS. At 6 months, the mean SBP fell by 6.1 mm Hg (to 127.0 mm Hg) in the intervention group and increased by 5.1 mm Hg (to 136.5 mm Hg) in the control group, resulting in an

adjusted between-group difference of -10.7 mm Hg (95% CI: -11.8 to -9.6 mm Hg; $P < 0.001$) (Table 2, Supplemental Tables 16 to 18). The ICC for SBP was 0.17. This substantial within-village correlation in BP levels indicates BP is shaped by shared community and household environments. Compared with baseline (Figure 2A), the distribution of SBP shifted leftward in the intervention group and rightward in the control group (Figure 2B). A BP level $<130/80$ mm Hg was achieved in 46.2% of the participants in the intervention group vs 23.6% of the participants in the control group (adjusted OR: 5.3, 95% CI: 4.3-6.6). After the withdrawal of the intervention, the mean difference in SBP between the intervention and control groups narrowed to -3.7 mm Hg (95% CI: -4.9 to -2.6), but remained statistically significant (Table 2, Supplemental Table 16, Figure 2C). The village-level changes in SBP from baseline to 6 months in the intervention and control groups are illustrated in Supplemental Figure 3. Across nearly all intervention villages, mean SBP decreased during follow-up, whereas most control

TABLE 1 Baseline Characteristics of Participants

	Healthy Family (n = 4,001)	Usual Care (n = 4,000)
Age, y	60.6 ± 9.1	61.8 ± 9.4
Female	2,068 (51.7)	2,043 (51.1)
Body mass index, kg/m ²	24.6 ± 3.5	24.3 ± 3.5
Primary school and below	1,401 (35.0)	1,630 (40.8)
Systolic blood pressure, mm Hg	133.0 ± 18.9	131.3 ± 18.1
Diastolic blood pressure, mm Hg	80.1 ± 10.3	78.7 ± 10.3
History of hypertension	1,444 (36.1)	1,232 (30.8)
History of diabetes mellitus	387 (9.7)	400 (10.0)
History of dyslipidemia	453 (11.3)	370 (9.3)
History of coronary artery disease	117 (2.9)	132 (3.3)
History of stroke	292 (7.3)	389 (9.7)
History of atrial fibrillation	27 (0.7)	19 (0.5)
History of heart failure	14 (0.3)	14 (0.4)
History of chronic kidney disease	12 (0.3)	14 (0.4)
History of chronic obstructive pulmonary disease	19 (0.5)	19 (0.5)
History of asthma	20 (0.5)	21 (0.5)
History of a malignant tumor	15 (0.4)	12 (0.3)
Current alcohol consumption	345 (8.6)	319 (8.0)
Current smoker	898 (22.4)	853 (21.3)
GPAQ scale score ^a	2,400 (600-5,280)	1,680 (0-5,040)
Mini-Eat scale score ^b	15.3 (12.9-18.3)	14.6 (11.9-17.7)

Values are mean ± SD, n (%), or median (IQR). Data are shown for the intention-to-treat population. ^aGlobal Physical Activity Questionnaire (GPAQ) scale calculates total metabolic equivalent of a task in minutes per week based on self-reported levels of different grades of activity: 8.0 × vigorous work (days per week) × vigorous work (minutes per day) + 4.0 × moderate work (days per week) × moderate work (minutes per day) + 4.0 × transport (days per week) × transport (minutes per day) + 8.0 × vigorous recreation (days per week) × vigorous recreation (minutes per day) + 4.0 × moderate recreation (days per week) × moderate recreation (minutes per day). ^bThe Mini-Eat scale is a short 9-item questionnaire to assess consumption of common food groups known to affect health.

villages showed stable or increased SBP levels over the same period. The intervention effect was consistent at both the village and family levels (Supplemental Tables 16, 19, and 20). Additionally, the intervention showed a significantly stronger effect in older adults, and in those with a history of hypertension, elevated SBP (≥140 mm Hg) and DBP (≥90 mm Hg), at baseline (Figure 3, Supplemental Table 21).

LIFESTYLE, BP MONITORING, AND ANTIHYPERTENSIVE DRUG THERAPY. The health instructors and family leaders showed a high level of fidelity in implementing the intervention (Supplemental Table 22), with 66.3% to 79.7% of participants uploading monthly BP measurements as instructed over 6 months (Supplemental Table 23). However, there was minimal between-group differences in diet, physical activity, and weight during the study period, as shown in Supplemental Tables 24 to 26. Individuals in the intervention group were more likely to use antihypertensive drugs (adjusted OR: 1.6, 95% CI: 1.1-2.3) (Table 2). Each class of

antihypertensive medication, except beta-blockers, was used by a significantly higher proportion of individuals in the intervention group compared with the control group (Supplemental Tables 27 and 28). During the study period, the BP monitoring platform was actively used, even after the support from the study group was withdrawn at 6 months (Supplemental Table 29).

SAFETY RESULTS. At 6 months, SAEs occurred in 112 (2.8%) participants in the intervention group and 91 (2.3%) participants in the control group (Table 2). Cardiovascular events were comparable in the intervention and control groups (28 events, 24 participants [0.6%] vs 21 events, 18 participants [0.5%]) and all-cause mortality was similarly (0.2%) in the groups (Supplemental Table 30). Injurious falls and biochemical abnormalities occurred in <0.2% of the intervention group.

COST OF INTERVENTION. The cost of implementing the multifaceted intervention in rural areas was 224.12 CNY (~USD \$31.20) per participant (Supplemental Table 31).

DISCUSSION

We have shown that a coordinated, multicomponent, community- and family-based intervention ‘left-shifted’ the population distribution of BP, with an adjusted between-group SBP reduction of 10.7 mm Hg over 6 months, in rural China. Although the between-group difference in SBP diminished after the active intervention was withdrawn, part of the initial reduction in SBP was maintained in the intervention group (from 133.2 mm Hg at baseline to 127.3 mm Hg at 12 months). The persistence of part of the initial benefit suggests that some intervention effects were maintained beyond the active intervention period. However, the attenuation of the treatment effect indicates that achieving sustained population-level BP reduction is likely to require ongoing implementation infrastructure rather than a time-limited intervention. The intervention was not associated with an apparent excess of SAEs or prespecified AESIs supporting the feasibility of implementing this community-based strategy at scale.

Although the intervention incorporated several lifestyle-focused components, changes in measured lifestyle-related indicators, including physical activity, dietary scores, and body weight, were relatively modest and may not fully explain the magnitude of BP reduction observed. The observed treatment effect, therefore, was likely mediated through multiple complementary mechanisms, including increased BP

TABLE 2 Primary and Secondary Efficacy and Safety Outcomes

	Baseline	Follow-Up	Change (mm Hg)	Least Squares Means Change (mm Hg, 95% CI) ^a	Adjusted Mean Difference (mm Hg, 95% CI) ^a
Primary outcome: change in SBP (mm Hg) from baseline to 6 months					
Health family (n = 3,892)	133.1 ± 18.9	127.0 ± 12.2	−6.1 ± 13.0	−5.3 (−6.0 to −4.5)	−10.7 (−11.8 to −9.6)
Usual care (n = 3,901)	131.4 ± 18.1	136.5 ± 16.4	5.1 ± 15.0	5.5 (4.6-6.3)	
Secondary outcomes					
Change in DBP (mm Hg) from baseline to 6 months					
Health family (n = 3,892)	80.1 ± 10.3	78.7 ± 10.3	−3.1 ± 7.4	−2.8 (−3.2 to −2.4)	−5.4 (−6.0 to −4.8)
Usual care (n = 3,901)	77.0 ± 8.0	81.3 ± 9.1	2.6 ± 9.0	2.6 (2.1-3.0)	
Change in SBP (mm Hg) from baseline to 12 months					
Health family (n = 3,617)	133.2 ± 19.1	127.3 ± 15.6	−5.9 ± 16.5	−4.7 (−5.5 to −3.9)	−3.7 (−4.9 to −2.6)
Usual care (n = 3,518)	131.5 ± 18.1	130.3 ± 16.3	−1.2 ± 17.1	−1.0 (−1.8 to −0.1)	
Participants with SBP <130 mm Hg and DBP <80 mm Hg in 6 months					
Health family (n = 3,892)	1,367 (35.1%)	1,800 (46.2%)	↘	↘	OR: 5.3 (4.3-6.6) ^b
Usual care (n = 3,901)	1,587 (40.7%)	922 (23.6%)	↘	↘	
Participants taking antihypertensive therapy at 6 months					
Health family (n = 3,892)	1,257 (32.3%)	1,238 (31.8%)	↘	↘	OR: 1.6 (1.1-2.3) ^b
Usual care (n = 3,901)	1,106 (28.4%)	1,007 (25.8%)	↘	↘	
Safety					
Serious adverse events during 6 months of follow-up ^c					
Health family (n = 3,892)	↘	112 (2.8%)	↘	↘	↘
Usual care (n = 3,901)	↘	91 (2.3%)	↘	↘	↘
Adverse events of special interest at 6 months of follow-up ^d					
Health family (n = 3,892)	↘	24 (0.6%)	↘	↘	↘
Usual care (n = 3,901)	↘	18 (0.5%)	↘	↘	↘

Values are mean ± SD or n (%) unless otherwise indicated. Data are shown for the intention-to-treat population. Multiplicity adjustment was planned using the sequential Holm-Sidak procedure. Given that all secondary outcomes demonstrated $P < 0.001$, all comparisons remained statistically significant after adjustment. ^aLeast-squares means within group and mean differences between groups are calculated by a mixed-effects regression model, with adjustment for township, age, sex, baseline SBP, and hypertension status as fixed effects, and village and family (nested within village) as random effects. ^bOR calculated in logistic regression model with the independent variable of interest, and adjusted for township, age, sex, baseline SBP and history of hypertension as fixed effects, and village and family (nested within village) as random effects. ^cSerious adverse events were prespecified to include events that may or may not be considered to be related to the treatment and that were life-threatening or resulted in hospitalization or prolongation of existing hospitalization, persistent or clinically significant disability or incapacity, medical or surgical intervention to prevent permanent impairment to bodily structure or function, or death. A patient could have more than 1 event. ^dIncludes hypotension, syncope, hyponatremia, hyperkalemia, injurious falls, and acute kidney injury.

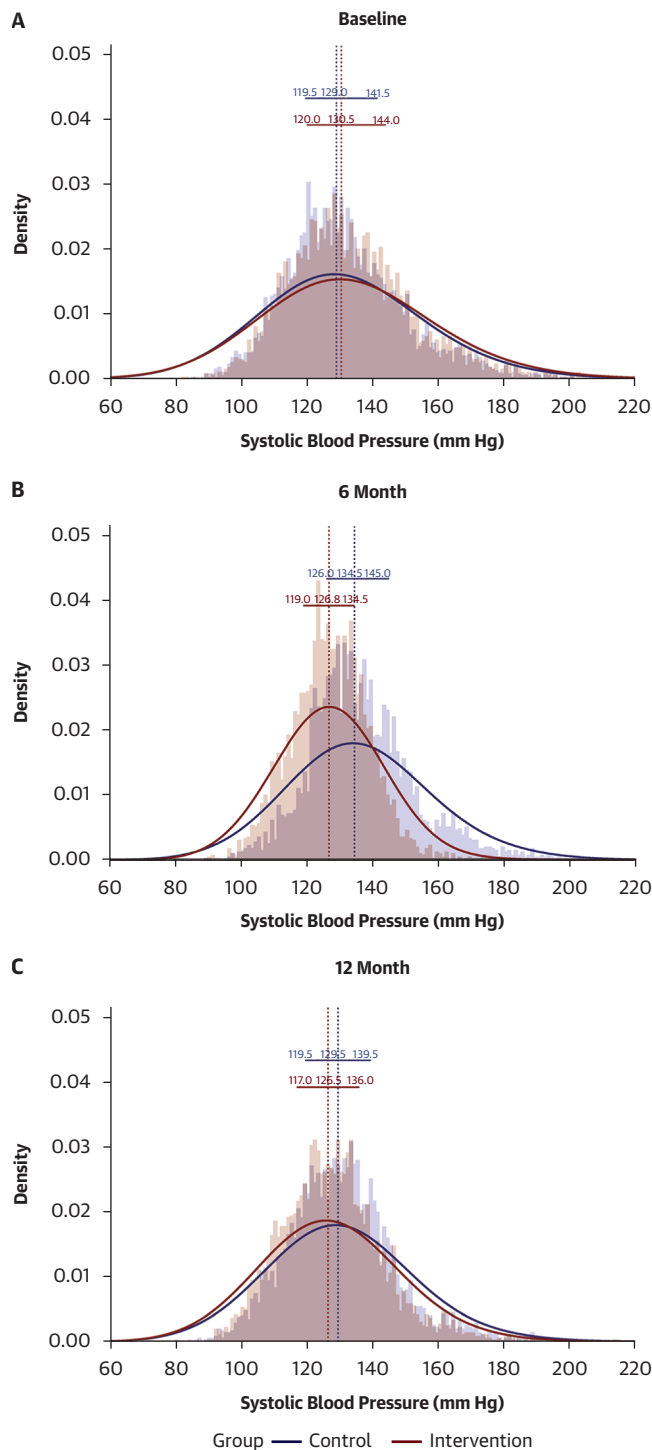
DBP = diastolic blood pressure; SBP = systolic blood pressure.

awareness through home monitoring, greater healthcare engagement, higher use of antihypertensive medications, salt substitution, and reinforcement of health behaviors within family and community networks. These findings suggest that the effectiveness of the intervention may have arisen not from lifestyle modification alone, but from the combined influence of behavioral, pharmacologic, monitoring, and social mechanisms operating simultaneously.

In the SsaSS (Salt Substitute and Stroke Study),¹⁸ use of a salt substitute (75% sodium chloride and 25% potassium chloride by mass) was associated with a between-group SBP difference of −3.34 mm Hg (95% CI: −4.51 to −2.18 mm Hg). Although the salt substitute formulation used in our study contained a

lower proportion of potassium chloride than that used in SsaSS, this component of the intervention likely contributed meaningfully to BP reduction.

Similarly, in the COBRA-BPS (Community-Based Intervention for Managing Hypertension in Rural South Asia) trial,¹⁹ the mean SBP reduction at 24 months was 5.2 mm Hg greater in the intervention group than in the control group (95% CI: 3.2-7.1). That intervention primarily involved home visits by trained government community health workers for BP monitoring, counselling, and care coordination. In contrast, our study adopted a broader family-based approach that engaged multiple family members and incorporated coordinated interventions targeting lifestyle behaviors, BP monitoring, medication adherence, and health care utilization. The more

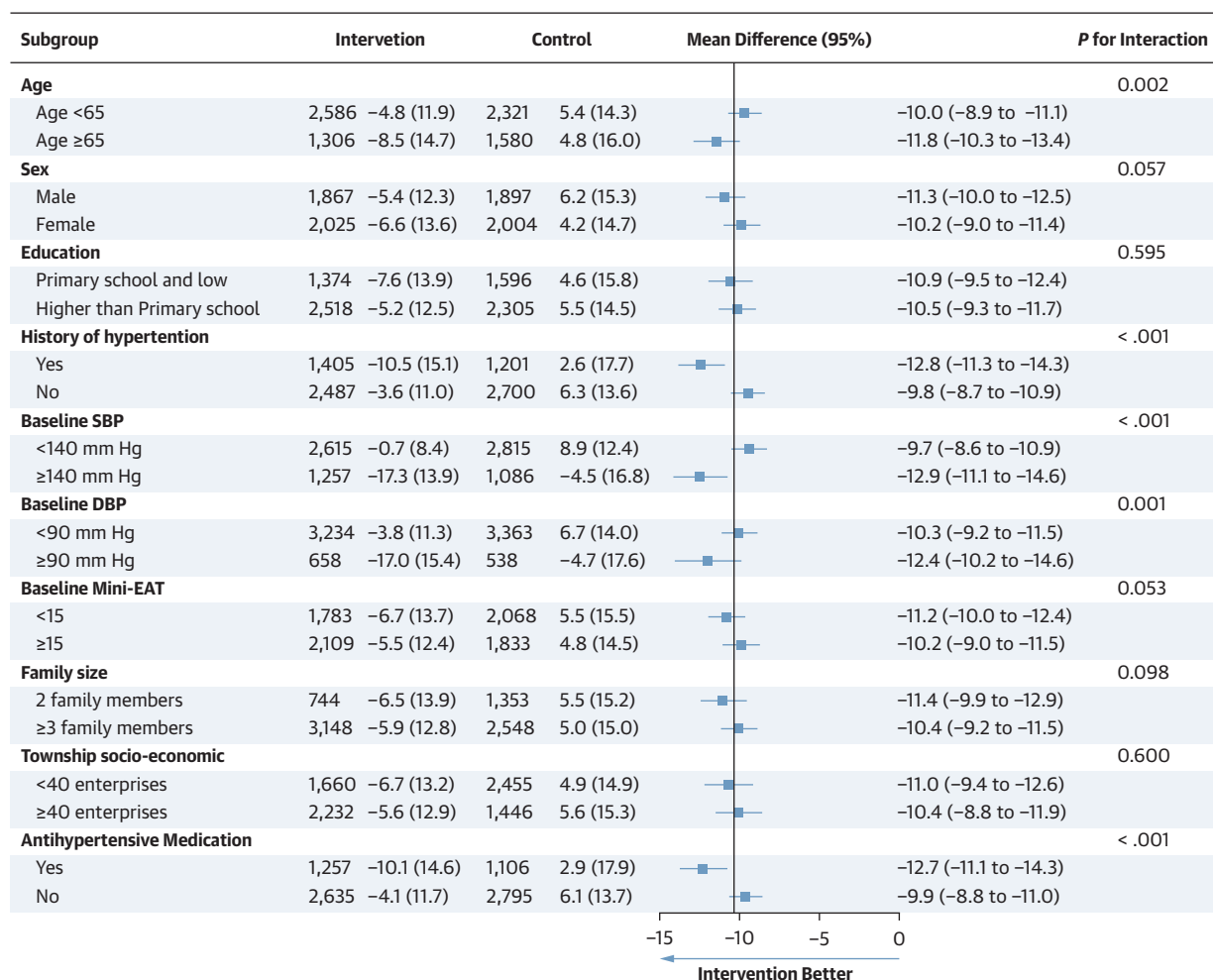
FIGURE 2 Distribution of Mean Systolic Blood Pressure at Baseline, 6 Months, and 12 Months

Distribution of mean systolic blood pressure (mm Hg) between the healthy family intervention group and the usual care (control) group at (A) baseline, (B) 6 months, and (C) 12 months. Solid curves represent kernel density estimates of systolic blood pressure distributions. Shaded areas indicate histogram densities. Vertical dashed lines represent medians, and horizontal whiskers indicate the interquartile range (25th–75th percentile). Numeric labels denote the 25th percentile, median, and 75th percentile values.

intensive involvement of family and community networks may therefore partly explain the larger reduction in SBP observed in our study. In the CRHCP (China Rural Hypertension Control Project) study,⁵ trained village doctors initiated and titrated antihypertensive medications according to a standardized treatment protocol. Village doctors also provided health coaching on home BP monitoring, lifestyle modification, and medication adherence, resulting in a between-group difference in mean SBP of 14.5 mm Hg (95% CI: –15.7 to –13.3 mm Hg). However, unlike our study, the CRHCP study enrolled only patients with diagnosed hypertension and a substantially higher baseline SBP (mean, 156 mm Hg), which may partly explain the larger BP reduction observed in that study.

The attenuation of the treatment effect between 6 and 12 months reinforces the challenges of sustaining population-level BP reduction over time. The partial attenuation observed after this period suggests that long-term maintenance of BP improvement may require sustained implementation infrastructure and continued support rather than a 1-time intervention. In our study, the digital platform remained active after several other intervention components were withdrawn, which may have contributed to maintaining part of the intervention effect during follow-up. Prior studies have similarly demonstrated that digital health approaches, including telemonitoring, smartphone-based interventions, and online self-management platforms, can support sustained BP reduction by facilitating ongoing engagement, self-monitoring, and treatment adherence.^{20–22} These findings suggest that sustainable population BP control may require a transition from short-term intervention delivery toward longer-term systems of reinforcement and monitoring embedded within community and primary care structures.

The transient elevation in mean SBP at 6 months in the control group, particularly among participants without hypertension, was likely driven by seasonal variation in BP. Seasonal fluctuation in BP is a well-documented physiological phenomenon.^{23–25} For instance, a meta-analysis reported that mean home SBP increases by approximately 6.1 mm Hg from summer to winter.²⁴ Consistent with this pattern, the control group in our study exhibited an increase in SBP from a baseline of 131.4 mm Hg to 136.5 mm Hg at 6 months, a net rise of 5.1 mm Hg, which falls within the expected range of seasonal variation. The SBP trajectories observed in the control group are consistent with findings from other rural hypertension trials,⁵ in which SBP levels in the control group also fluctuated over time, with fluctuations observed

FIGURE 3 Change in Systolic Blood Pressure (mm Hg) Over 6 Months by Different Characteristics of Participants and Villages

Forest plot showing subgroup analyses of systolic blood pressure reduction between intervention and control groups. Estimates are presented as mean differences with 95% CIs. Negative values favor the intervention group, indicating greater reductions in systolic blood pressure. P values for interaction assess heterogeneity of treatment effects across subgroups. The vertical line represents the overall adjusted mean difference.

across 6-month intervals. Supporting this hypothesis, SBP in the control group returned to baseline levels at 12 months, when data were collected during the same season as the baseline measurements, reinforcing the seasonal influence on BP. However, participants in both study groups were assessed during the same seasonal periods within a narrow time window at each follow-up visit. Therefore, although seasonal variation may have influenced absolute SBP levels over time, it is unlikely to have materially biased the between-group comparison of the intervention effect. Moreover, incorporating actual visit week as a time-varying covariate in the longitudinal model did not change the results materially.

In China, high SBP accounted for over 25 million disability-adjusted life years lost due to stroke in 2019.²⁶ Despite substantial efforts to strengthen primary healthcare since 2009, evidenced by a 10-fold increase in government financial support from USD \$2.8 billion at that time to USD \$33.7 billion in 2019,²⁷ the impact on hypertension control has been modest. In the most recent national survey conducted from 2021 to 2022, awareness, treatment, and control of hypertension remained at 43.3%, 38.7%, and 12.9%, respectively.²⁸ As a consequence, the Chinese government launched the “Healthy Family” campaign in early 2024, a nationwide initiative aimed at promoting individual responsibility for health through education on essential health knowledge, skills, and

ecofriendly lifestyle choices. Our study exemplifies how comprehensive policy integration and the use of family-centered, community-based interventions can play a crucial role in addressing hypertension and, potentially, associated cardiovascular risks at the population level.

While implementation mechanisms are necessarily context-dependent, we believe the core principles underlying our intervention are potentially transferable to other health care settings. Several components of the intervention are not unique to rural China but instead represent broadly applicable population BP control strategies. These include home BP monitoring, which may improve BP awareness and thereby influence health behaviors, treatment adherence, and health care engagement²¹; family engagement, which is globally relevant because medication adherence, dietary habits, salt intake, and health care utilization are strongly shaped by household environments²⁹; and the integration of prevention strategies into existing community structures, which may provide a scalable approach for population-level cardiovascular prevention. More broadly, our findings reinforce the concept that coordinated multilevel strategies, rather than isolated interventions, may be necessary to achieve meaningful population BP reduction.^{4,7} At the same time, certain aspects of implementation may be context-dependent. The effectiveness of the intervention may have been facilitated by the established village doctor network, strong family and community social structures, and the predominance of home-prepared diets in rural China. Adaptation to other settings, therefore, may require different implementation models depending on local healthcare infrastructure, workforce availability, cultural norms, and patterns of health care utilization.

STUDY LIMITATIONS. As its conduct was limited to Ruyang County, the findings may not be generalizable to other regions, although data indicate the population has good representation of other rural areas in China. Although the cluster-randomized design did not include predefined intervillage distance criteria, there is a low likelihood of contamination, given the relative geographical isolation of Ruyang and the exclusive distribution of intervention materials. We acknowledge that measurement error, absence of ambulatory BP monitoring, and regression to the mean may have influenced the measurements of BP taken by multiple staff and at

different time points. Finally, as the study was not powered to detect differences in cardiovascular outcomes, further research is required to determine whether the intervention can reduce cardiovascular events as a primary outcome.

CONCLUSIONS

This cluster-randomized trial demonstrated that a coordinated, multicomponent, family- and community-based intervention produced a substantial reduction in BP at the population level in a real-world rural setting. These findings support the feasibility of shifting cardiovascular prevention beyond the clinic, suggesting that community-based systems of care can meaningfully reduce BP at the population level.

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APPENDIX For supplemental information, tables, and figures, please see the online version of this paper.